ECE 3317

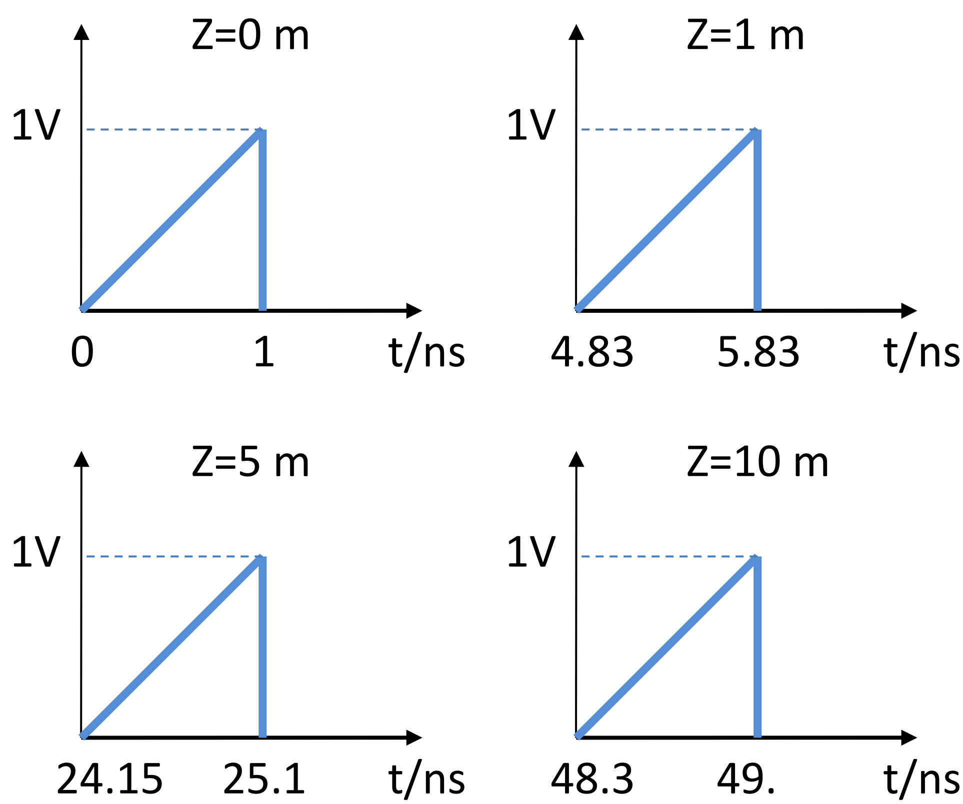
Homework #3

Answers

1) A coaxial cable is filled with lossless Teflon, having a relative permittivity of 2.1. (Teflon is nonmagnetic, so the relative permeability is unity.) How long with it take a signal to travel a one-meter length of the cable?

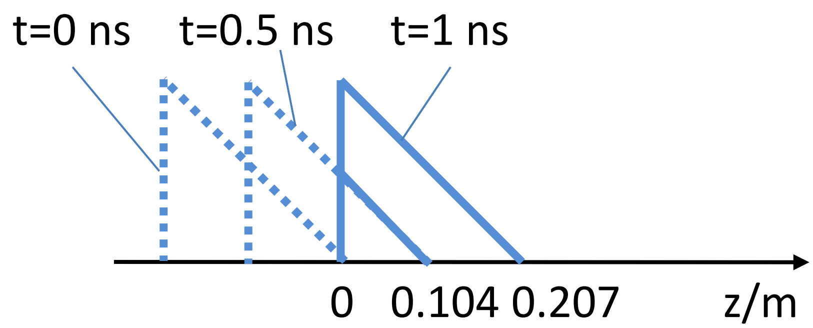
   4.83 ns

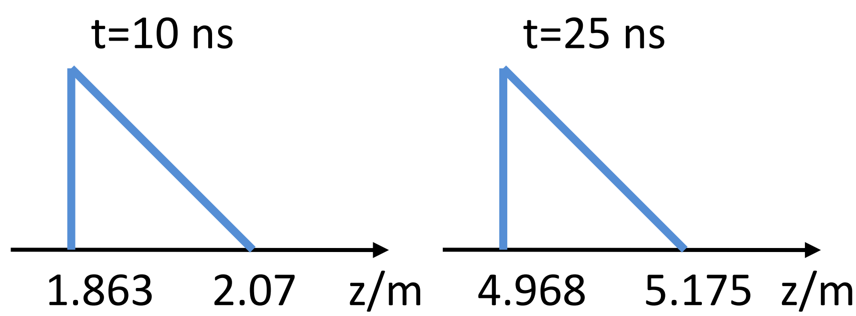
2) A coaxial cable is 10 meters long, and is filled with lossless Teflon, having a relative permittivity of 2.1. There is a matched load at the end of the line. At the input of the line, a sawtooth waveform is applied, having a peak voltage of 1.0 [V] and a duration of 1.0 [ns] (the time from the beginning of the waveform until the end). Make a plot (versus time) of what an oscilloscope would read if it were connected to the line at various locations, corresponding to the following values of *z*: 0.0 [m], 1.0 [m], 5 [m], 10 [m]. Plot out to 50 [ns] in your plot.

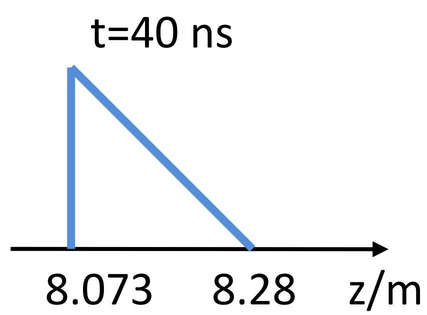


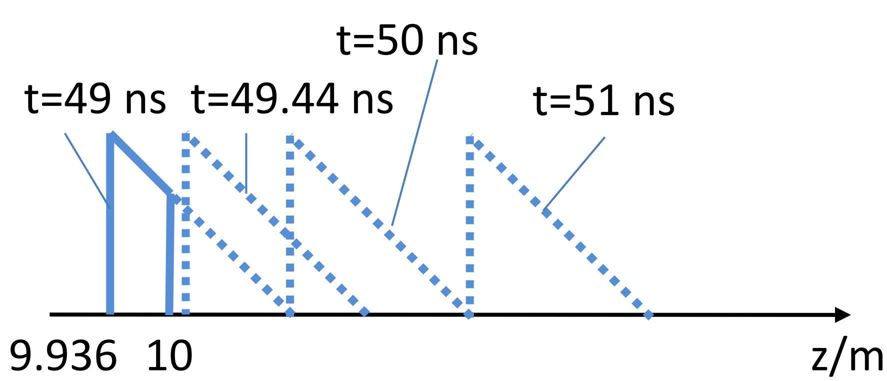
3) A coaxial cable is 10 meters long, and is filled with lossless Teflon, having a relative permittivity of 2.1. There is a matched load at the end of the line. At the input of the line, a sawtooth waveform is applied, having a peak voltage of 1.0 [V] and a duration of 1.0 [ns] (the time from the beginning of the waveform until the end). Make a plot of the voltage as a

function of distance z on the line, at the following times: 0.0 [ns], 0.5 [ns], 1.0 [ns], 10 [ns], 25 [ns], 40[ns], 49 [ns], 49.44 [ns], 50 [ns], 51 [ns]. For each time, choose appropriate limits on your horizontal axis (distance z) to make the plot easy to read.









4) A coaxial cable for TV applications has an outer diameter of 3 [mm]. The coax is filled with lossless Teflon, having a relative permittivity of 2.1. What is the necessary diameter of the

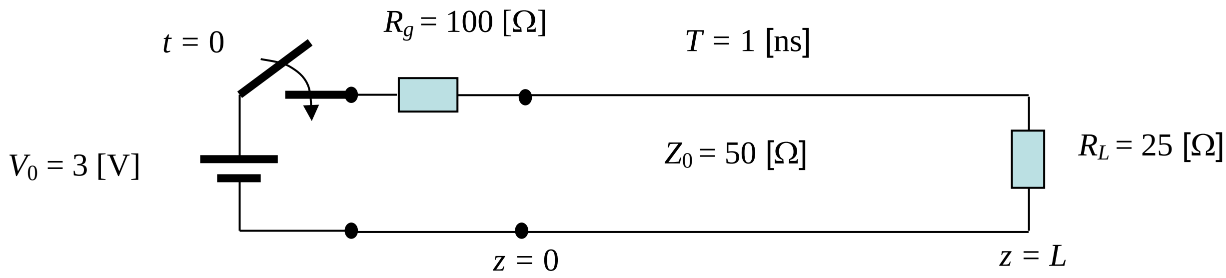
inner conductor (wire) in order for the characteristic impedance of the coax to be 75 [Ω]?

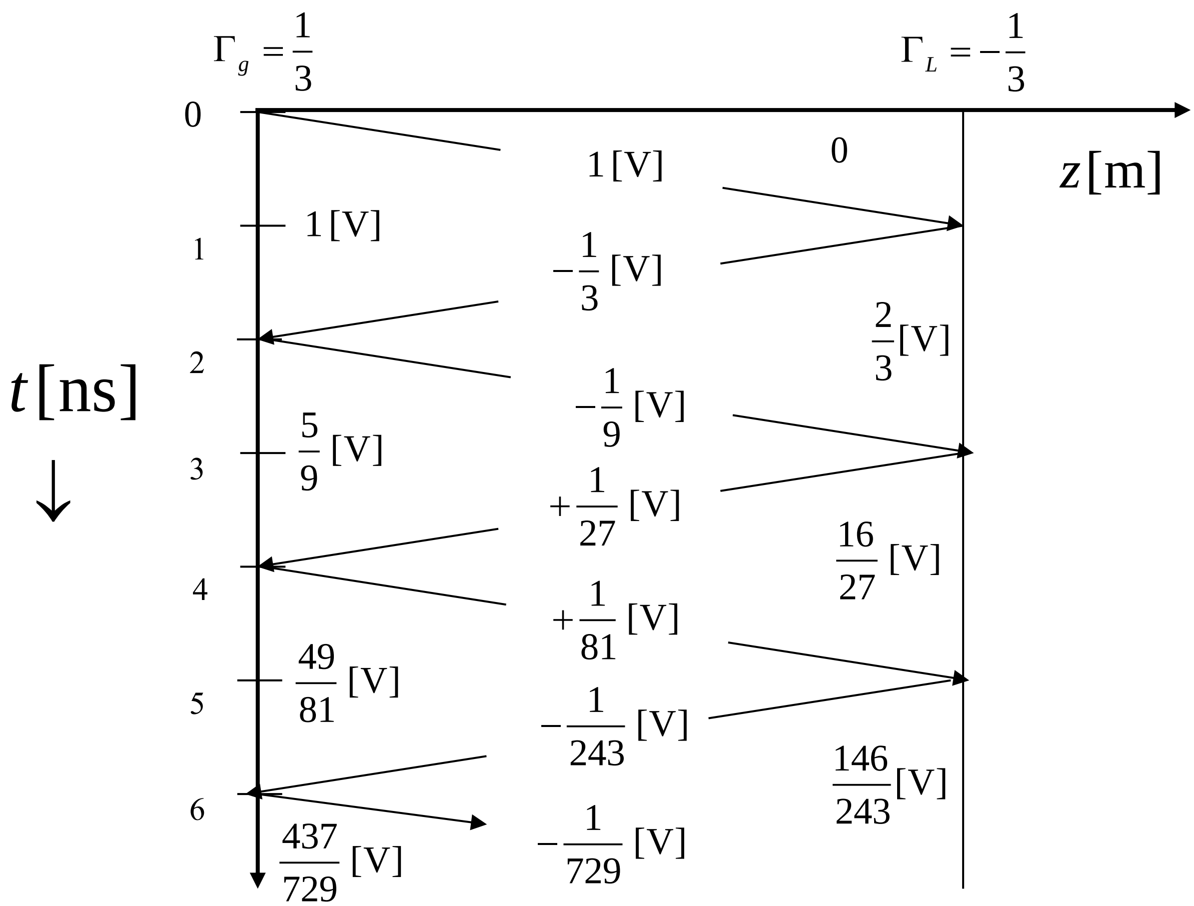
0.490 mm

5) A twin line transmission line for TV applications is supposed to have a characteristic impedance of 300 [Ω]. If the spacing (center-to-center) between the two wires is 8 [mm], what should be the diameter of the wires, assuming that they are in air? Use both the approximate formula for the characteristic impedance (having the natural logarithm function) and the exact one (involving the inverse hyperbolic cosine function) and compare the results.

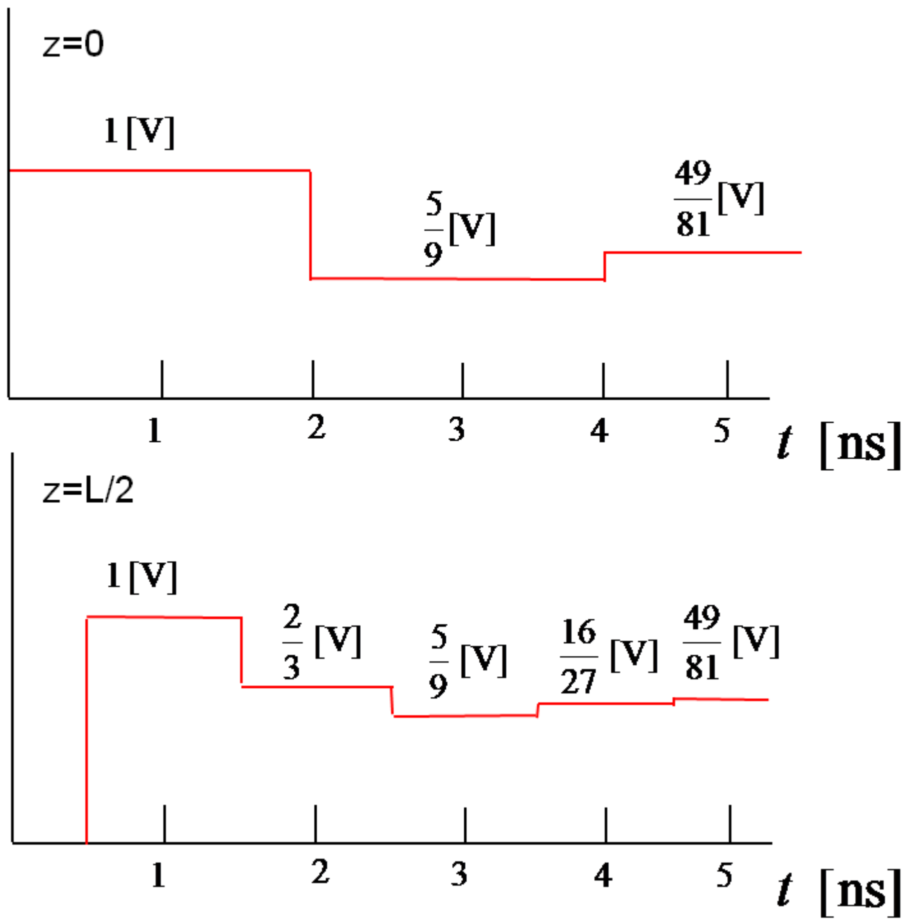
Approximate: 1.313 mm
Exact: 1.305 mm

6) Consider the problem shown below. Construct a bounce diagram for this problem.

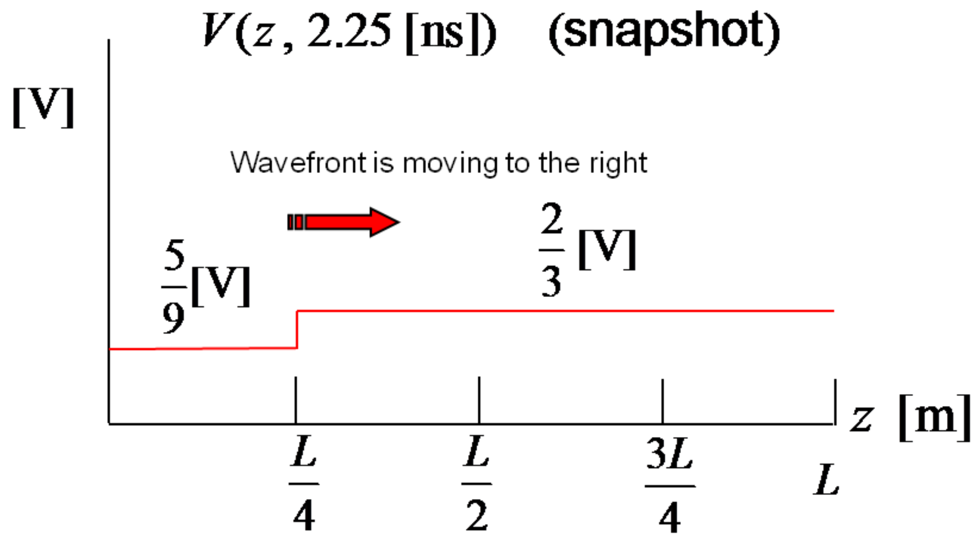




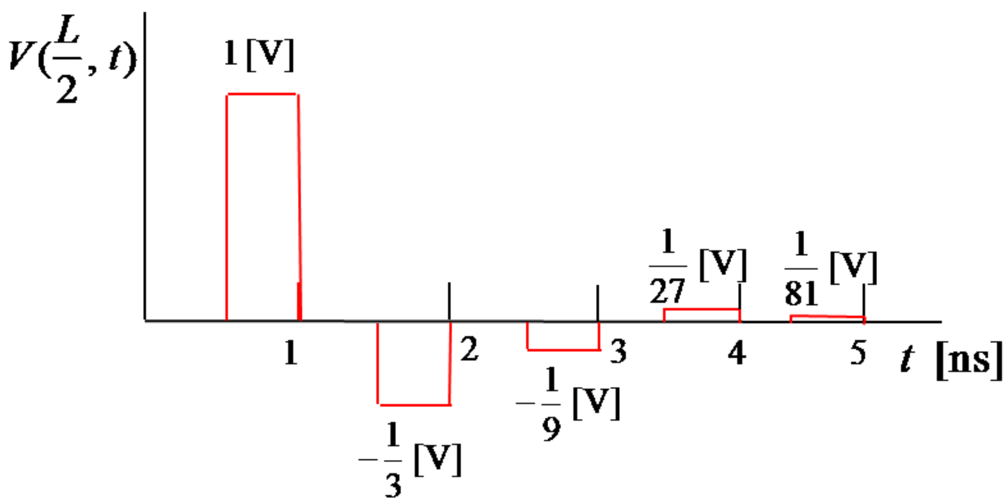
7) For the same problem as above, use the bounce diagram to make an accurate oscilloscope trace if the oscilloscope were attached to the input end at z = 0. Repeat for z = L/2.



8) For the same problem as above, use the bounce diagram to make a snapshot of the voltage on the line at the time t = 2.25 [ns]. Put an arrow on your diagram to indicate which way the wave front is traveling.



9) For the same problem as above, now assume that we have a digital pulse of amplitude 3.0 [V] and duration W = 0.5 [ns] applied at the input. Use the bounce diagram to construct an accurate oscilloscope trace for an oscilloscope connected at $z = L/2$.



10) For the same problem as above (with the digital pulse), use the bounce diagram to make an accurate snapshot of the voltage on the line at t = 0.75 [ns]. Repeat for t = 1.25 [ns].